Of course, there could be interesting things going on at higher energies, so by including a cutoff we're admitting that we're not getting exactly the right answer. But Wilson showed that what we do get is generally more than good enough. We can precisely characterize how, and roughly by how much, any new high-energy phenomena could possibly affect the low-energy world we actually see. By admitting our ignorance in this way, what we're left with is an *effective field theory*—one that doesn't presume to be an exact description of anything, but one that can successfully fit the data we actually have. Modern quantum field theorists recognize that all of their best models are actually effective field theories.

This leaves us with a good news/bad news situation. The good news is that we are able to say an enormous amount about the behavior of particles at low energies, using the magic of effective field theory, even if we don't know everything (or anything) about what's happening at higher energies. We don't need to know all the final answers in order to say something reliable and true. That's a big part of why we can be confident that the laws of physics governing the particles and forces that make up you and me and our everyday environments are completely known: those laws take the form of an effective field theory. There's plenty of room to discover new particles and forces, but either they must be too massive (high energy) to have yet been produced in experiments, or they interact with us so incredibly weakly that they can't possibly have an effect on tables and chairs and cats and dogs and other pieces of the architecture of our low-energy world.

The bad news is that we would very much like to learn more about what's really going on at high energies and short distances, but the magic of effective field theory makes that extremely hard. It's good that we can accurately describe low-energy physics no matter what is going on at higher energies, but it's also frustrating because this seems to imply that we can't infer what's going on up there without somehow probing it directly. This is why particle physicists are so enamored of building ever larger and higher-energy particle accelerators; that's the only reliable way we know of to discover how the universe works at very small distances.